Yeon Seonwoo

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RESEARCH INTERESTS

My research focuses on how intelligent systems acquire, connect, and utilize knowledge to reason and act in the real world. I am interested in the relationship between knowledge, reasoning, and planning, spanning topics such as continual learning, retrieval, agentic search, and large language models. My recent work explores how these capabilities can be integrated into practical systems, where real-world constraints such as latency, scalability, and reliability fundamentally influence the design of intelligent agents.

KEYWORDS

Large Language Models (LLMs), LLM Pre-training, Inference Optimization, Agentic Search, Retrieval-Augmented Generation (RAG), Query Understanding

EXPERIENCE

Applied Scientist, Amazon (Manhattan, NY) Jul 2023 - Current

Knowledge Integration for Foundation Models

- Led development of 1B–8B domain-specialized language models for shopping and e-commerce applications.
- Developed continual pre-training methodologies that improve domain knowledge integration while preserving general-purpose capabilities.
- Enabled broad adoption of domain-specialized language models across Amazon shopping applications.

Entity-Grounded Query Understanding for Search

- Enabled real-time query understanding with mid-sized language models for search applications through inference optimization techniques including multi-token prediction.
- Developed entity-grounded query understanding by aligning language-model reasoning with Prime Video catalog knowledge.
- Developed finite-state-machine constrained decoding with jump-forward execution for efficient catalog-grounded generation.

Applied Scientist Intern, Amazon (Bellevue, WA) Mar 2022 - Jun 2022

Developed an unsupervised sentence representation learning method to more effectively capture semantic similarity between texts.

Research Scientist Intern, Adobe Research Sep 2021 - Dec 2021

Developed word- and relation-level semantic graph representations of documents for domain-specific document retrieval.

EDUCATION

KAIST Daejeon, Republic of Korea

Ph.D. Student in School of Computing Sep 2017 - Aug 2022

Advisor: Alice Oh

Thesis title: “Distant-Supervision for Question Answering”

KAIST Daejeon, Republic of Korea

M.S. in School of Computing Mar 2016 - Aug 2017

Advisor: Alice Oh

Thesis title: “Event Prediction Using Vector Representation in Hawkes Processes”

Sungkyunkwan University Suwon, Republic of Korea

B.S. in Computer Science & Engineering Mar 2012 - Feb 2016

AWARDS

Naver Ph.D. Fellowship, 2020

PUBLICATION

Juhee Son, **Yeon Seonwoo**, Alice Oh, James Thorne, Seunghyun Yoon, "Multi-hop Database Reasoning with Virtual Knowledge Graph", *Workshop on Knowledge Graphs and Large Language Models 2024.*

Chani Jung, Dongkwan Kim, Jiho Jin, Jiseon Kim, **Yeon Seonwoo**, Yejin Choi, Alice Oh, Hyunwoo Kim, "Perceptions to Beliefs: Exploring Precursory Inferences for Theory of Mind in Large Language Models", *EMNLP 2024.*

Yeon Seonwoo, Guoyin Wang, Sajal Choudhary, Changmin Seo, Jiwei Li, Xiang Li, Puyang Xu, Sunghyun Park, Alice Oh. "Ranking-Enhanced Unsupervised Sentence Representation Learning." *ACL 2023*

Yeon Seonwoo, Seunghyun Yoon, Franck Dernoncourt, Trung Bui, Alice Oh. "Virtual Knowledge Graph Construction for Zero-Shot Domain-Specific Document Retrieval." *COLING 2022*

Changyoon Lee, **Yeon Seonwoo**, and Alice Oh. "CS1QA: A Dataset for Assisting Code-based Question Answering in an Introductory Programming Course." *NAACL 2022*

Yeon Seonwoo*, Juhee Son*, Jiho Jin, Sang-Woo Lee, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. "Two-Step Question Retrieval for Open-Domain QA." *ACL-Findings. 2022.*
*Contributed equally

Yeon Seonwoo, Sang-Woo Lee, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. "Weakly Supervised Pre-Training for Multi-Hop Retriever." *ACL-Findings. 2021.*
Code URL: `https://github.com/yeonsw/LOUVRE`

Yeon Seonwoo, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. "Context-Aware Answer Extraction in Question Answering." *EMNLP (long). 2020.*
Code URL: `https://github.com/yeonsw/BLANC`

Yeon Seonwoo, Sungjoon Park, Dongkwan Kim, and Alice Oh. "Additive Compositionality of Word Vectors." *Workshop on Noisy User-generated Text at EMNLP. 2019.*
Code URL: `https://github.com/yeonsw/OLIVE`

Sungjoon Park, **Yeon Seonwoo**, Jiseon Kim, Jooyeon Kim and Alice Oh. "Denoising Recurrent Neural Networks for Classifying Crash-Related Events." *IEEE Transactions on Intelligent Transportation Systems. 2019.*

Yeon Seonwoo, Sungjoon Park, and Alice Oh. "Hierarchical Dirichlet Gaussian Marked Hawkes Process for Narrative Reconstruction in Continuous Time Domain." *EMNLP (long). 2018.*

INVITED TALKS

Semantic Alignment at Various Context Sizes
BrainLink: State, Limitations, and Future of Large Language Models (LLM), South Korea — Oct. 26, 2022

Weakly-Supervised Pre-Training for Multi-Hop Retrieval
KISTI, South Korea — Feb. 08, 2022
Naver, South Korea — Dec. 22, 2021
LG AI Research, South Korea — Nov. 22, 2021

Context-Aware Answer Extraction in Question Answering
Naver, South Korea — Jul. 01, 2020

Additive Compositionality of Word Vectors
Naver, South Korea — Dec. 14, 2019

Hierarchical Dirichlet Gaussian Marked Hawkes Process for Narrative Re-

construction in Continuous Time Domain
NCSoft AI, South Korea Jan. 25, 2019

ACADEMIC SERVICE

Reviewer
ACL Rolling Review (Oct 2021, Nov 2021, Jan 2022, Apr 2022, June 2022, July 2022, Sep 2022), NeurIPS2022, EMNLP 2022, ICLR 2022, NeurIPS 2021, EMNLP 2021, ACL 2021, ICLR 2021

SCHOLARSHIP

Korea National Science&Technology Scholarship (Mar 2012 - Feb 2016)

REFERENCES

Prof.Alice Oh, School of Computing, KAIST, alice.oh@kaist.edu